

CS 311: Algorithm Design and Analysis — Lecture 1

Introduction and Asymptotic Performance — Study Questions

Batch 1 - Questions 1 to 12: The Course

Q1. What is the stated purpose of the course?

- a) An introduction to database administration
- b) A rigorous introduction to the design and analysis of algorithms
- c) A hands-on introduction to a programming language
- d) A review of discrete mathematics proofs

Answer: b) A rigorous introduction to the design and analysis of algorithms

Q2. According to the first slide, the course is NOT:

- a) A course that uses a textbook
- b) A lab or programming course, and not a math course either
- c) A course about algorithms
- d) A course with an exam

Answer: b) A lab or programming course, and not a math course either

Q3. Which textbook is used in the course?

- a) The Art of Computer Programming by Knuth
- b) Introduction to Algorithms by Cormen, Leiserson, Rivest, and Stein (3rd edition)
- c) Data Structures and Algorithms in Java by Goodrich
- d) Algorithms by Sedgewick and Wayne (4th edition)

Answer: b) Introduction to Algorithms by Cormen, Leiserson, Rivest, and Stein (3rd edition)

Q4. How is the textbook described in the slides?

- a) A book only for the teaching assistant
- b) An optional book that is rarely needed
- c) An excellent reference you should own
- d) A book replaced entirely by the slides

Answer: c) An excellent reference you should own

Q5. What percentage of the grade is class work?

- a) 10%
- b) 40%
- c) 25%
- d) 50%

Answer: c) 25%

Q6. What percentage of the grade is the midterm exam?

- a) 25%
- b) 10%
- c) 30%
- d) 50%

Answer: a) 25%

Q7. What percentage of the grade is the final exam?

- a) 25%
- b) 60%

- c) 40%
- d) 50%

Answer: d) 50%

Q8. What is the prerequisite of the course?

- a) CS 211 Data Structures and Algorithms
- b) CS 311 Operating Systems
- c) CS 101 Introduction to Programming
- d) CS 411 Theory of Computation

Answer: a) CS 211 Data Structures and Algorithms

Q9. Which of the following is a stated learning outcome of the course?

- a) Learning to administer computer networks
- b) Learning the main classic algorithms in various domains
- c) Learning to build graphical user interfaces
- d) Learning to design relational databases

Answer: b) Learning the main classic algorithms in various domains

Q10. Which of the following is a stated learning outcome of the course?

- a) Learning to write operating system kernels
- b) Learning to manage software projects
- c) Learning techniques for designing efficient algorithms
- d) Learning hardware circuit design

Answer: c) Learning techniques for designing efficient algorithms

Q11. Which of the following is a stated learning outcome of the course?

- a) Avoiding the use of design techniques
- b) Studying only theoretical proofs with no problem solving
- c) Memorizing the textbook word for word
- d) Applying the algorithms and design techniques to solve problems

Answer: d) Applying the algorithms and design techniques to solve problems

Q12. Which of the following is a stated learning outcome of the course?

- a) Knowing the history of programming languages
- b) Knowing only the syntax of pseudocode
- c) Having a sense of the complexities of various problems in different domains
- d) Knowing the prices of different computer systems

Answer: c) Having a sense of the complexities of various problems in different domains

Batch 2 - Questions 13 to 22: Asymptotic Performance and Notation

Q13. In this course, what do we care about most?

- a) The exact running time on one specific computer
- b) Asymptotic performance
- c) The length of the source code
- d) The programming language used

Answer: b) Asymptotic performance

Q14. Asymptotic performance asks:

- a) How many programmers are needed to write it?
- b) How many lines of code does the algorithm have?

- c) How does the algorithm behave as the problem size gets very large?
- d) How fast is the algorithm on a tiny input?

Answer: c) How does the algorithm behave as the problem size gets very large?

Q15. Which two resources are mentioned as being analyzed asymptotically?

- a) Running time and memory/storage requirements
- b) Disk color and screen size
- c) Memory and network bandwidth only
- d) Running time and electricity cost

Answer: a) Running time and memory/storage requirements

Q16. The slides say that by now you should have an intuitive feel for which notation?

- a) Roman numerals
- b) Binary-tree notation
- c) Matrix notation
- d) Asymptotic (big-O) notation

Answer: d) Asymptotic (big-O) notation

Q17. Which running times does the slide use as examples when asking what big-O notation means?

- a) $O(1)$, $O(\log n)$, and $O(2^n)$
- b) $O(n)$, $O(n^2)$, and $O(n \lg n)$
- c) $O(n!)$, $O(n^n)$, and $O(\sqrt{n})$
- d) $O(n^3)$, $O(n^4)$, and $O(n^5)$

Answer: b) $O(n)$, $O(n^2)$, and $O(n \lg n)$

Q18. What question about big-O is posed for the student to think about?

- a) How does asymptotic running time relate to asymptotic memory usage?
- b) How does big-O relate to the color of the monitor?
- c) How does big-O relate to the price of a computer?
- d) How does big-O relate to the number of lines of code?

Answer: a) How does asymptotic running time relate to asymptotic memory usage?

Q19. What is the first task stated in the notation section?

- a) To design a user interface
- b) To define asymptotic notation more formally and completely
- c) To memorize the textbook index
- d) To install the software for the course

Answer: b) To define asymptotic notation more formally and completely

Q20. Analysis of algorithms is performed with respect to:

- a) A specific brand of computer
- b) A specific programming language only
- c) A random guess
- d) A computational model

Answer: d) A computational model

Q21. Which computational model will usually be used?

- a) A distributed multi-processor cluster
- b) A quantum computer
- c) A GPU-only machine

d) A generic uniprocessor random-access machine (RAM)

Answer: d) A generic uniprocessor random-access machine (RAM)

Q22. What does RAM stand for in this context (the model)?

- a) Read-only access memory
- b) Random-access machine
- c) Recursive array machine
- d) Rapid algorithm model

Answer: b) Random-access machine

Batch 3 - Questions 23 to 38: The RAM Model, Input Size, Running Time, and Cases

Q23. In the RAM model, how expensive is it to access memory?

- a) All memory is equally expensive to access
- b) Memory cost grows with the address
- c) Disk access is cheaper than RAM access
- d) Registers are cheaper than main memory

Answer: a) All memory is equally expensive to access

Q24. Which of the following does the RAM model assume about concurrency?

- a) Unlimited concurrent operations
- b) No concurrent operations
- c) Concurrent operations only for sorting
- d) Exactly two concurrent operations

Answer: b) No concurrent operations

Q25. In the RAM model, how long do all reasonable instructions take?

- a) A random amount of time
- b) Time proportional to the input size
- c) Unit time
- d) Time proportional to the number of bits

Answer: c) Unit time

Q26. Which kind of operation is explicitly an exception to the 'unit time' assumption?

- a) Function calls
- b) Addition
- c) Assignment
- d) Comparison

Answer: a) Function calls

Q27. What does the RAM model assume about word size?

- a) Word size is unlimited
- b) Word size is always one bit
- c) Word size grows with the input size
- d) Constant word size, unless we are explicitly manipulating bits

Answer: d) Constant word size, unless we are explicitly manipulating bits

Q28. Which of the following is NOT an assumption of the RAM model?

- a) Memory access time depends on the memory location
- b) All reasonable instructions take unit time
- c) There are no concurrent operations

d) All memory is equally expensive to access

Answer: a) Memory access time depends on the memory location

Q29. Time and space complexity are generally a function of:

a) The color of the output

b) The input size

c) The language's syntax

d) The programmer's experience

Answer: b) The input size

Q30. For sorting, how is input size characterized?

a) By the number of loops in the code

b) By the total number of bits

c) By the number of nodes and edges

d) By the number of input items

Answer: d) By the number of input items

Q31. For multiplication, how is input size characterized?

a) By the number of nodes and edges

b) By the total number of bits

c) By the number of input items

d) By the number of function calls

Answer: b) By the total number of bits

Q32. For graph algorithms, how is input size characterized?

a) By the number of input items only

b) By the number of colors

c) By the total number of bits only

d) By the number of nodes and edges

Answer: d) By the number of nodes and edges

Q33. In the lecture, running time is defined as:

a) The number of lines in the source code

b) The amount of memory used

c) The number of seconds on a stopwatch

d) The number of primitive steps that are executed

Answer: d) The number of primitive steps that are executed

Q34. Except for function calls, most statements roughly require:

a) The same amount of time

b) No time at all

c) An amount of time that depends on the line number

d) Exponentially more time each time they run

Answer: a) The same amount of time

Q35. Which of the following is given as an example of a statement taking roughly constant time?

a) Sorting an array

b) Searching a graph

c) $y = m * x + b$

d) Calling a recursive sort on a large array

Answer: c) $y = m * x + b$

Q36. What does worst-case analysis provide?

- a) The expected running time over random inputs
- b) A lower bound on running time
- c) An upper bound on running time, which is an absolute guarantee
- d) No information about running time

Answer: c) An upper bound on running time, which is an absolute guarantee

Q37. What does average-case analysis provide?

- a) The best possible running time
- b) An absolute guarantee on running time
- c) The expected running time
- d) The memory used

Answer: c) The expected running time

Q38. Why should average-case analysis be 'treated with care', according to the slides?

- a) Because it ignores the input size
- b) Because it always gives a wrong answer
- c) Because it only works for sorting
- d) Because it is unclear what 'average' means (random equally likely inputs vs. real-life inputs)

Answer: d) Because it is unclear what 'average' means (random equally likely inputs vs. real-life inputs)

Essay Questions

Q39. Describe the RAM (random-access machine) model used for analyzing algorithms in this course.

Answer:

The analysis is performed with respect to a computational model; usually a generic uniprocessor random-access machine (RAM) is used:

- All memory is equally expensive to access.
- There are no concurrent operations.
- All reasonable instructions take unit time (except, of course, function calls).
- Constant word size, unless we are explicitly manipulating bits.

Q40. Explain how input size is characterized for different problems, and how running time is measured.

Answer:

Input size: time and space complexity are generally a function of the input size, and how it is characterized depends on the problem:

- Sorting: the number of input items.
- Multiplication: the total number of bits.
- Graph algorithms: the number of nodes and edges.

Running time: the number of primitive steps that are executed. Except for the time of executing a function call, most statements (e.g., $y = m \cdot x + b$, $c = 5/9 \cdot (t - 32)$, $z = f(x) + g(y)$) roughly require the same amount of time. We can be more exact if need be.

Q41. Compare worst-case and average-case analysis.

Answer:

Worst case: provides an upper bound on running time — an absolute guarantee.

Average case: provides the expected running time. It is very useful but must be treated with care, because it is not obvious what 'average' means: random (equally likely) inputs, or real-life inputs.